

Question:

Carla is downloading a 200 GB file. Normally she can download 2 GB/minute, but 40% of the way through the download, Windows forces a restart to install updates, which takes 20 minutes. Then Carla has to restart the download from the beginning. How long does it take to download the file?

Reasoning + Answer:

<think> To solve this, first calculate the normal download time, then add the time taken for the Windows update restart, and finally, add the time for the download to start again from the beginning. The file size is 200 GB. At 2 GB / minute, it would normally take $200 / 2 = 100$ minutes to download the entire file. 40% of the way through is $0.4 * 200 = 80$ GB. Downloading 80 GB at 2 GB / minute takes $80 / 2 = 40$ minutes. After the restart, she has to start the 200 GB download again, which takes another 100 minutes. Adding the 20 minutes for the Windows update restart, the total time is $40 + 20 + 100 = 160$ minutes. </think><answer>160</answer>

